AgentOS Studio - Architecture Blueprint (Version 1)

Purpose

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Core Domains

Core Domains 1: Workspace Manager, Agent Runtime, Workflow Engine, Memory Engine, Tool Registry, MCP Gateway, Evaluation Engine, Prompt Registry, Secrets Manager, Artifact Store, Execution Graph Store, Telemetry Service, RBAC, Marketplace, Deployment Manager, SDK, CLI, Desktop UI and Web UI. The architecture follows clean boundaries, asynchronous messaging, deterministic state transitions, immutable execution logs, checkpointing, replay, dependency injection, provider abstraction, policy-driven authorization, audit trails, plugin isolation, semantic versioning, extensible APIs, OpenTelemetry tracing, distributed scheduling, execution snapshots, prompt versioning, evaluation datasets, benchmark pipelines, human approval gates, multi-model routing, cost governance, and disaster recovery. Every subsystem exposes REST, WebSocket and SDK interfaces where appropriate. Engineers should prioritize composability, testability, scalability, backward compatibility, and developer experience over framework lock-in.

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Future Chapters

Recommended future chapters include: detailed sequence diagrams, ER diagrams, API specifications, protobuf schemas, plugin lifecycle, deployment topologies, Kubernetes manifests, Helm charts, CRDs, database schemas, cache invalidation strategy, message contracts, benchmark methodology, SDK reference, UI wireframes, design tokens, accessibility, localization, roadmap, contribution guide, release process, governance model, threat model, penetration testing checklist, compliance mapping, SLA definitions, and performance targets.